

Jerome Agapay

Software Engineer

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PROFESSIONAL SUMMARY

Full-stack developer with 7+ years of experience, specializing in back-end development with JavaScript and Node.js. Strong analytical skills with a focus on cloud technologies and AWS solutions. Recently building local-first RAG (Retrieval-Augmented Generation) systems in Python. Known for quickly learning new technologies and adapting to different business requirements across various industries.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, PHP 7.x

Frontend Technologies: ReactJS, Redux, React Native, jQuery, Tailwind CSS, Ant Design, NextJS

Backend Technologies: Node.js, Express.js, Serverless Framework, Redis, Laravel, FastAPI

Databases: PostgreSQL, MySQL, MariaDB, Aurora Serverless v2, Liquibase, SQLite

AI/RAG: LlamaIndex, Ollama, RAG pipelines, vector search (Qdrant, ChromaDB), hybrid retrieval & reranking, embeddings, RAGAS evals

Cloud/AWS: Lambda, API Gateway, RDS, S3, S3 Glacier, CloudFront, Cognito, CloudWatch, SQS, EC2, Amplify, CDK

AI Tools: Claude Code, Anthropic API, AI-assisted development, LLM integration

Development Tools: Git, Docker, CI/CD pipelines, Agile/Scrum methodologies

Operating Systems: Linux (Ubuntu), macOS, Windows

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (SAA-C03), Oct 2025

Credly Badge: <https://www.credly.com/badges/785dceab-7921-4e55-92f0-bc7a2daea3a7>

PROFESSIONAL EXPERIENCE

Senior Software Engineer | [HealthyLife Company Pty Ltd](#) | Jul 2022 - May 2024

E-commerce and digital health & wellness platform

- Fixed UI bugs including user interaction issues and data display problems across the platform.
- Built new UI features while maintaining and improving existing legacy codebase.
- Ensured mobile responsiveness across the website for optimal user experience on all devices.
- Worked with legacy code challenges including jQuery and older CSS, successfully modernizing components.

Technologies: HTML, CSS, jQuery, SASS, Tailwind CSS, PHP, Laravel

Senior Software Engineer | [PayMaya Philippines Inc.](#) | May 2021 - Jul 2022

Business management platform for payment, banking, and loan solutions

- Built CRUD operations using AWS Lambda and Serverless Framework for both frontend (React) and backend (Node.js).
- Integrated SQS to handle file processing for disbursement feature - users upload Excel files with thousands of rows for money distribution.
- Performed load testing on SQS and Lambda integration to optimize upload times and improve processing speed.
- Participated in code reviews and bi-weekly sprint planning with development team, QA, and business stakeholders.

Technologies: React 17, Redux, ES6, Node.js, AWS Lambda, SQS, RDS, S3, Serverless Framework

Tech Lead | [BotBros AI, Rocket Labs Consulting Inc.](#) | Jun 2017 - Jan 2021

Airship Logistics Management System

Courier tracking and delivery platform for logistics businesses

- Led full-stack development of logistics management system handling bookings, agent management, and cash flow operations.
- Facilitated Agile processes including sprint planning, daily stand-ups, retrospectives, and code reviews.
- Researched and implemented AWS cloud solutions and emerging technologies for scalable architecture decisions.
- Collaborated with stakeholders during project planning to define requirements and optimize development workflows.

Achievement: Won Top 3 position in IdeaSpace Acceleration Program for innovative logistics solution.

Technologies: ReactJS, Redux, Node.js, Serverless, AWS Lambda, API Gateway, RDS, Cognito, CloudWatch, CloudFront, S3

PeopleServe - Human Resource Information System

Recruitment and staffing solutions platform

- Collaborated with clients during initial planning phases to ensure clear requirements and realistic project timelines.
- Provided technical mentorship to interns and junior developers, improving team onboarding and skill development.
- Delivered post-deployment support including troubleshooting and technical issue resolution.

Technologies: Laravel 5.2, ReactJS, Redux, MariaDB, MySQL 5.6

Additional Projects:

- Entrego Booking Website: Developed responsive homepage, booking system, and reusable React components in collaboration with Zalora team.
- Chatbot Solutions: Built Facebook Messenger and Slack bots with system integrations using Botkit JS and Node.js.
- Company Website: Designed and developed brand-focused website using Ghost CMS.

QA Intern | Ideyatech, Inc. | Oct 2016 - Jan 2017

G-Force Dance Center - Reservation and booking platform

- Designed and implemented comprehensive test cases, improving test coverage for web application features.
- Performed manual testing of website functionality, identifying bugs and ensuring optimal user experience.

PERSONAL PROJECTS

PH Law RAG - Retrieval-Augmented Generation Assistant | Apr 2026 - Present

Local-first RAG assistant over Philippine law primary sources, running entirely on a local LLM

- Built a hybrid retrieval pipeline combining dense vector search (Qdrant) with sparse BM25, followed by cross-encoder reranking for higher answer relevance.
- Implemented incremental document sync via content hashing so re-runs skip unchanged statutes.
- Delivered grounded generation that cites article/section numbers and abstains when the corpus can't support an answer.
- Measured answer quality with RAGAS semantic eval metrics; shipped a Streamlit chat UI and a FastAPI service.

Technologies: Python, LlamaIndex, Ollama (Mistral), Qdrant, BM25, cross-encoder reranking, RAGAS, FastAPI, Streamlit, SQLite, Docker

AWS & Amazon Bedrock Docs RAG Lab - Documentation Q&A System | Apr 2026

Local-first RAG system over curated AWS and Amazon Bedrock documentation with grounded citations

- Built a document ingestion pipeline that fetches an allowlist of AWS and Amazon Bedrock pages, normalizes HTML to clean text, and detects changes via content hashing to only reprocess what changed.
- Curated Bedrock documentation covering model inference, Knowledge Bases, Guardrails, Agents, Flows, batch inference, and evaluations as the primary RAG corpus.
- Chunked and embedded documents locally with sentence-transformers and stored vectors in ChromaDB, tracking metadata and versions in SQLite.
- Exposed the pipeline through both a Typer CLI and a FastAPI service, with an eval harness scoring answers against a question dataset.

Technologies: Python, Ollama, sentence-transformers, ChromaDB, FastAPI, Typer, SQLite, BeautifulSoup, Amazon Bedrock docs

PSILO - Personal Cloud Storage | Feb 2026 - Apr 2026

Self-hosted image storage platform built as an alternative to commercial cloud storage solutions

- Architected and built a full-stack personal cloud storage app using AWS CDK for infrastructure-as-code, provisioning Cognito, API Gateway, Lambda, SQS, Aurora Serverless v2, and S3 Glacier Flexible Retrieval.
- Implemented cost-optimized cold storage using S3 Glacier Flexible Retrieval, significantly reducing hosting costs compared to standard S3.
- Designed monorepo structure separating frontend (Next.js), infrastructure (CDK stacks), and Lambda services for clean architecture and maintainability.
- Applied Backend for Frontends (BFF) pattern with decoupled async photo metadata processing via SQS and Dead Letter Queue for reliability.
- Used Aurora Data API with Lambda (no VPC/NAT gateway) to simplify database connectivity and reduce infrastructure complexity.
- Integrated Drizzle ORM with Aurora Serverless v2 with database migrations for type-safe SQL and version-controlled schema management.

Technologies: Next.js, TypeScript, Node.js, AWS CDK, Lambda, API Gateway, Aurora Serverless v2, S3 Glacier, Cognito, SQS, Drizzle ORM

Inventory Management System | May 2025 - Present

Client project - Digital inventory system replacing Excel-based workflows

- Built inventory management system for client to replace error-prone Excel-based tracking, preventing data duplication and loss.
- Implemented RPC approach instead of REST since no public APIs were needed.
- Applied clean architecture principles to improve code organization and maintainability.
- Deployed on Vercel for reliable hosting and easy client access.

Technologies: Next.js, TypeScript, Clean Architecture, RPC, Vercel

EDUCATION
